ECE 3340

Numerical Methods

Exam 2A

Name:____

Random Numbers _________ / 30

Partial Pivoting _________ / 35

Matrix Conditioning _________ / 35

Total _________ / 100

Random Numbers. Find the maximum period of a 4-bit MLCG with a modulus of 15 and a multiplier of 2. **[15 points]**

$\boldsymbol{x_0}$ **must be co-prime with** $\boldsymbol{m}$**:**

Random Numbers. Calculate the parameters for an LCG that will provide the maximum period for a 3-bit random number generator. Show each value in the sequence starting with $x_0 = 1$. **[15 points]**

$\boldsymbol{m} =$

$\boldsymbol{c} =$

$\boldsymbol{a} =$

Partial Pivoting and Determinants. Calculate the determinant $|A|$ using partial pivoting. Fill in the blank values at each step. If a vector or matrix doesn't change from one step to the next, you don't have to fill it in (just mark it as the same). **[30 points]**

$$|\boldsymbol{A}| = \begin{vmatrix} \mathbf{0} & \mathbf{1} & \mathbf{5} & \mathbf{7} \\ \mathbf{0} & \mathbf{2} & \mathbf{4} & \mathbf{2} \\ \mathbf{1} & \mathbf{3} & \mathbf{2} & \mathbf{2} \\ \mathbf{0} & \mathbf{1} & \mathbf{3} & \mathbf{5} \end{vmatrix} =$$

$$\begin{bmatrix} & & & \\ & & & \\ & & & \\ & & & \end{bmatrix} \qquad \boldsymbol{\ell} = \begin{bmatrix} \\ \\ \\ \\ \end{bmatrix}$$

$$\begin{bmatrix} & & & \\ & & & \\ & & & \\ & & & \end{bmatrix} \qquad \boldsymbol{\ell} = \begin{bmatrix} \\ \\ \\ \\ \end{bmatrix}$$

$$\begin{bmatrix} & & & \\ & & & \\ & & & \\ & & & \end{bmatrix} \qquad \boldsymbol{\ell} = \begin{bmatrix} \\ \\ \\ \\ \end{bmatrix}$$

$$\begin{bmatrix} & & & \\ & & & \\ & & & \\ & & & \end{bmatrix} \qquad \boldsymbol{\ell} = \begin{bmatrix} \\ \\ \\ \\ \end{bmatrix}$$

$$\begin{bmatrix} & & & \\ & & & \\ & & & \\ & & & \end{bmatrix} \qquad \boldsymbol{\ell} = \begin{bmatrix} \\ \\ \\ \\ \end{bmatrix}$$

What is $|A^{-1}|$? What formula do you use to calculate this? **[5 points]**

LU Decomposition and Inversion. Calculate the missing column of the matrix inverse given the LU decomposition of a matrix $\boldsymbol{A}$:

$$\boldsymbol{L} = \begin{bmatrix} 1 & 0 & 0 \\ 2 & 2 & 0 \\ 3 & 1 & 2 \end{bmatrix} \qquad \boldsymbol{U} = \begin{bmatrix} 1 & 2 & 1 \\ 0 & 3 & 2 \\ 0 & 0 & -2 \end{bmatrix}$$

and **show your work**. **[20 points]:**

$$\begin{bmatrix} {}^{11}\!/_{6} & -{}^{7}\!/_{24} & \\ -{}^{2}\!/_{3} & {}^{1}\!/_{12} & \\ {}^{1}\!/_{2} & {}^{1}\!/_{8} & \end{bmatrix}$$

You can check the answer with your calculator. Estimate the condition number for $\boldsymbol{A}$ using the $\|\ \|_1$ and $\|\ \|_\infty$ norms **[15 points]**.